

COGNITION

- COGNITION**
 - EXPERIMENTAL: AUTOMATIC, NOT AWARE OF DOING SOMETHING (RIDING A BIKE)
 - REFLECTIVE: CONTROL COGNITIVE PROCESS, WE REFLECT (LINE WAITING A REPORT)
- RAISED**
 - FAST: 2+2
 - SLOW: 21x29 WE NEED TO THINK ABOUT THE PROCESS

- DIVIDE COGNITION AMONG DIFFERENT COGNITIVE PROCESS (SOME ARE CONSCIOUS SOME AUTOMATIC)

COGNITION CAN

- UNDERSTAND CAPABILITIES AND LIMITATIONS OF INDIVIDUALS
- ALLOW TO 'PREDICT' AND UNDERSTAND ERRORS
- PROVIDE THEORIES AND USEFUL MODELING TOOLS FOR DESIGN OF BETTER INTELLIGENT PRODUCTS

PERCEPTION

- PSYCHOLOGICAL PROCESS THROUGH WHICH WE CAN INTERPRET SENSORY STIMULATION

- WHAT WE SEE ≠ WHAT WE INTERPRET

- ↳ USE LEGIBLE TEXT AND DISTINGUISHABLE ICONS

- **WELLER**: PEOPLE TOOK LESS TIME TO LOCATE ITEMS FOR INFORMATION THAT WAS GROUPED

COLORS

- WE DISTINGUISH MORE EASILY COLORS THAT CAUSE STRONG SIGNAL IN COLOR PERCEPTION CHANNELS

- RED-GREEN, YELLOW-BLUE, BLACK-WHITE

- DISTINCTIVE COLORS ARE BETTER

- AVOID STRONG OPPONENTS PAIRS

- AVOID COMBINATIONS HARD FOR COLOR-BLIND

GESTALT THEORY

- ↳ PERCEIVE OBJECTS AS WELL ORGANIZE PATTERNS RATHER THAN SEPARATE ITEMS

- **PROXIMITY**: GROUP TOGETHER SHAPES IF THEY ARE CLOSE TO EACH OTHER

- **SIMILARITY**: GROUP TOGETHER SIMILAR ELEMENTS REGARDLESS OF THEIR PROXIMITY (COLOR, SHAPE, SIZE)

- **CONTINUITY**: CREATE CONTINUOUS FIGURES (MAINLY LINES)

- **CLOSURE**: PERCEIVE INCOMPLETE OBJECTS AS COMPLETE + CLOSE AND FILL GAPS

DESIGN IMPLICATION

- ICONS TO DISTINGUISH MEANING READILY
- SPACE AND BORDERS TO GROUP INFORMATION
- AVOID AMBIGUITY

- BE CONSISTENT (W/ SHAPE, COLORS, FONTS)

- UNDERSTAND COMBS

PERCEPTION CAN BE BIASED BY

- EXPERIENCE → EXAMPLE BUILDINGS + FAMILIAR PATTERNS
- CONTEXT → "THE" VS "CAT"
- GOALS

- **DISTRIBUTED COGNITION**: FRAMEWORK AND METHOD TO EXAMINE INTERACTION BETWEEN PEOPLE AND OBJECTS BY MINIMIZING COMPLEX INTERDEPENDENCES BETWEEN PEOPLE AND OBJECTS + INFO. IS TRANSFORMED THROUGH DIFFERENT MEDIA

ATTENTION

- ABILITY TO DETECT AND RESPOND TO STIMULI
- HOW WE PROCESS INFORMATION IN ENVIRONMENT

- ALLOW TO BE SELECTIVE BUT LIMITS ABILITY TO KEEP TRACK OF ALL EVENTS

EXAMPLE HOTEL COST + SPACING ON SCREEN

- TULLIS. W/OUT SPACING 5.3 TO FIND COST WHILE W/ SPACING 3.2 SECONDS
- SAME INFORMATION DENSITY
- SECOND SCREEN GROUPED VERTICAL → 0

HEAVY VS LIGHT MULTITASKING

- H. → MORE PHONE DISTRACTION + CAN NOT FILTER IRRELEVANT INFORMATION

- **LOTTERIDGE**: H. EASILY DISTRACTED BUT ACHIEVE TASK IF DISTRACTING SOURCE IS RELEVANT TO THE TASK + IRRELEVANT SOURCE OF DISTRACTION PERFORM NEGATIVELY

DRIVING + PHONE

- LONGER TIME REACTION WHEN TALKING ON PHONE
- DRIVERS TENDS TO IMAGINE THE OTHER PERSON FACE WHEN TALKING → 2
- DRIVER CAN COOPERATE W/ FRONT SEAT TO DETECT RISK

- MULTITASKING CAN BE DETRIMENTAL SINCE IT OVERLOADS CAPACITY TO FOCUS

MEMORY

- ABILITY TO RETAIN INFO. OVER TIME

THREE STAGES:

- **ENCODING**: PLACE PIECE OF INFO. IN MEMORY

- **STORAGE**: WHEN INFO. IS STORED IN MEMORY

- **RETRIEVAL**: WHEN INFO. IS RECOVERED FROM STORAGE

THREE STORAGES:

- **SENSORY S.**: LARGE AND TRANSIENT MEMORY STORAGE, LARGE NUMBER OF INFO. BUT LIMITED AMOUNT OF TIME

- **SHORT-TERM S.**: WHAT YOU ARE CONSCIOUS OF, DECAY IN 20 S

- **LONG-TERM S.**: UNLIMITED STORAGE OF INFO. GENERALLY AVAILABLE TO US

- **MILLER**: 5-TO-9 ITEMS RECALL IN ORDER, NOT ALWAYS THE CASE, USED IN INTERACTION DESIGN → IN A BULLET LIST WE DO NOT DO RECALL BUT RECOGNITION

- **BYRNE-BOVARI**: COGNITIVE ERRORS ATM, POST-COMPLETION ERRORS WHEN WITHDRAW CASH BUT FORGOT CARD, THIS TEND TO HAPPEN WHEN USER HAVE ADDITIONAL STEP TO PERFORM AFTER PRIMARY TASK AND WHEN LOAD OF W.A. WAS HIGH. → REDESIGN ATMS

- DESIGNERS NEED TO REDUCE COGNITIVE LOAD BY AVOIDING LONG AND COMPLICATED PROCEDURES FOR CARRYING OUT TASK

- **WORKING MEMORY**: SYSTEM RESPONSIBLE FOR TEMPORARY STORAGE AND PROCESSING INFO. HELPS IN EVERYDAY LIFE, BUT HAVE SMALL CAPACITY SO IT CAN BE **OVERLOADED**, IT CONTAINS:

- **CENTRAL EXECUTIVE**: COORDINATES ACTIVITY

- **VISUAL-SPATIAL SKETCHPAD**: STORAGE AND MAINTENANCE OF VISUAL-SPATIAL INFO.

- **PHONOLOGICAL LOOP**: SHORT-TERM RETENTION

- **EPISODIC BUFFER**: LIMITED CAPACITY STORAGE, LINK WM. TO LONG TERM M.

- **RECOGNITION**: RECOGNIZE PREVIOUSLY SEEN STIMULI FROM NEW ONES ← EASIER

- **RECALL**: BRING TO PRESENT PAST STIMULI

- USE PICTURES (ICONS) WHEN POSSIBLE TO CONVEY FUNCTION

- **EXTERNAL COGNITION**: ARTIFACTS AND OBJECT AS AN EXTENSION OF OUR MIND. KNOWLEDGE NOT CONFINED IN THE BRAIN

- **COGNITIVE OFFLOADING**: STRATEGY TO AVOID FORGETTING AND EFFORT OF REMEMBERING USE DIARIES, NOTES, ... REMIND US WHAT WE NEED TO DO AND WHEN.

- **COGNITIVE FRAMEWORK**: EXPLAIN AND PREDICT USER BEHAVIOR WHEN INTERACTIVE W/ TECHNOLOGY

MENTAL MODELS:

- PEOPLE DEVELOP AN UNDERSTANDING OF A SYSTEM THROUGH LEARNING ABOUT AND USING IT

- HOW TO USE A SYSTEM + WHAT TO DO IN UNFAMILIAR SITUATIONS

- PEOPLE MAKE INFERENCE USING M.M. ON HOW TO CARRY TASKS

- INTERNAL CONSTRUCTION OF SOME ASPECTS OF EXTERNAL WORLD

- USE M.M. TO REASON ON A SYSTEM + HOW TO INTERACT W/ IT

- ARE INCOMPLETE, CONSTANTLY EVOLVING, NOT ACCURATE REPRESENTATION 2

- USERS 'SEE' THE SYSTEM W/ MENTAL MODELS (DESIGNER ALSO HAVE M.M.)

- **SYSTEM IMAGE**: SPECIFIC FUNCTIONING OF THE SYSTEM ITSELF

- NEED TO REDUCE GAP BETWEEN: USER M.M. + DESIGNER M.M. + SYSTEM IMAGE

- **VISIBILITY**: MORE VISIBLE FUNCTION → MORE USERS WILL USE IT, MAKE RELEVANT PART VISIBLE + WHAT NEED TO BE DONE OBVIOUS

- **FEEDBACK**: SENDING INFO. BACK TO USER (INCLUDE SOUND, ANIMATION, ...)

- **CONSTRAINTS**: RESTRICT POSSIBLE ACTIONS THAT CAN BE PERFORMED CAN BE PHYSICAL (SHAPE PREVENTS CERTAIN ACTION), CULTURAL (GRAVITY) OR LOGICAL (RELIES ON COMMON SENSE)

- **MAPPING**: RELATIONSHIP BETWEEN CONTROLS AND THE RESULT IN THE WORLD

- **CONSISTENCY**: INTERFACE THAT USE SIMILAR ELEMENTS FOR SIMILAR TASK

- INTERNAL C.: OPERATIONS BEHAVE SAME WITHIN AN APPLICATION

- EXTERNAL C.: OPERATIONS SAME ACROSS APPLICATIONS AND DEVICE

- **AFFORDANCES**: FUNCTIONAL ATTRIBUTES OF AN OBJECT THAT ALLOW PEOPLE TO KNOW HOW TO USE IT.

GULFS

- **EXECUTION**: BETWEEN USER AND SYSTEM - WHAT'S THE SYSTEM CURRENT STATE?

- **EVALUATION**: BETWEEN SYSTEM AND USER - HOW DO I USE THE SYSTEM?

- BRIDGING GULFS → REDUCE COGNITIVE EFFORT REQUIRED TO PERFORM TASK

- **EMBODIED INTERACTION**: INCORPORATE OUR RELATIONSHIP W/ THE WORLD AROUND US INTO THE DESIGN AND USE OF INTERACTIVE SYSTEM

- USER COMMUNICATE MEANING THROUGH THE INTERACTION W/ THE SYSTEM

- **EMBODIED METAPHORS**: UNDERSTAND ABSTRACT CONCEPTS IN CONCRETE CONCEPTS

Emotions

- FULL BODY, MIND BEHAVIOR RESPONSE TO A STIMULATION

- **EMOTION DESIGN**: DESIGN PRODUCTS TO EVOKE CERTAIN EMOTION RESPONSES IN USERS

- **EMOTIONAL AI OR AFFECTIVE COMPUTING**: SYSTEMS TO DETECT AND RECOGNISE SOMEONE EMOTIONS FROM FACIAL EXPRESSIONS, BODY MOVEMENTS, ...

- EMOTIONS GENERATE AFFECTIVE EXPERIENCES, TRIGGER A SERIES OF INTERNAL AUTOMATIC NEUROUS SYSTEMS ADJUSTMENTS, TRIGGER BEHAVIOURS THAT ARE OFTEN BUT NOT ALWAYS EXPRESSIVE

- **MOOD STATE**: WHEN AN EMOTION (THAT IS SHORT-LIVED AND AROUSED BY A SPECIFIC EVENT) IS REPEATED

- **RUSSEL**: VALENCE/AROUSAL, EMOTION IN 2D PLANE

- **VALENCE**: ATTRACTION VS DISLIKING → BABY FACE BIAS (NAIVE)

- **AROUSAL**: DEACTIVATION VS ACTIVATION

- **Ekman**: 6 BASIC EMOTIONS (ANGER, DISGUST, FEAR, HAPPINESS, SADNESS, SURPRISE + NEUTRAL)

- EMOTIONS IN 1980s TO INDICATE REBOOTING 22

- **NORMAN**: 3 LEVELS OF COGNITIVE PROCESS THAT GIVE RISE TO EMOTIONAL ASSOCIATIONS BETWEEN HUMAN USER AND PRODUCT

- **VISCERAL LEVEL**: SENSORY REACTION MEDIATED BY PHYSICAL CHARACTERISTIC OF THE INTERACTIVE SYSTEM. MAKE PRODUCTS LOOK, FEEL AND SOUND GOOD

- **BEHAVIORAL LEVEL**: EVALUATE DESIGN ACCORDING WHETHER IT ACHIEVES THE GOALS OF EFFECTIVENESS, EFFICIENCY AND SATISFACTION. FOCUS ON USABILITY AND PERFORMANCE.

- **REFLEXIVE LEVEL**: USER EVALUATE BENEFITS, CONSIDER THE MEANING AND PERSONAL VALUE OF A SYSTEM

- **MILMAN HIERARCHY OF NEEDS**: WHEN ONE NEED IS SATISFIED THE NEXT ONE BECOME DOMINANT

- **PHYSIOLOGICAL**: AIR, WATER, FOOD SLEEP

- **SAFETY**: PHYSICAL SAFETY, FINANCIAL SECURITY

- **SOCIAL**: LOVE, FRIENDSHIP SOCIETY

- **ESTEEM**: ACHIEVEMENT, SUCCESS, RESPECT BY OTHERS

- **SELF-ACTUALIZATION**: FULL POTENTIAL + BETTER VERSION

- **PICARD AFFECTIVE COMPUTING**: DESIGN SYSTEM CAPABLE OF RECOGNISE, INTERPRET AND RESPOND TO HUMAN EMOTIONS

- **AFFECTIVE SENSING**: SYSTEM THAT CAN RECOGNIZE EMOTIONS BY COLLECTING DATA THROUGH SENSORS + ALGORITHMS RECOGNISE ON

- **STEPS**: BASELINE DATA ACQUISITION → TASK-RELATED DATA ACQUISITION → DATA CLEANING → FEATURE SELECTION → TRAINING → VALIDATION → TESTING → FEEDBACK RESTRICTION

- **FACS**: MEASURE FACIAL ACTIVITY USING OBJECTIVE COMPONENTS FACIAL MOTIONS OR FACIAL ACTIONS TO IDENTIFY EMOTIONS

- IDENTIFY A SET OF ACTION UNITS (AUs) THAT IDENTIFY INDEP. MOTIONS OF THE FACE. TRAIN HUMANS TO DECOMPOSE AN EXPRESSION IN A SET OF AUs